

Client Integration Document
For
Parichay
(Single Sign-On)



Submitted by



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DEFINITIONS, ACRONYMS & ABBREVIATIONS

Terms	Meaning
Service Name	Name of the client service (client application)
Client Token / Local Token Id	The identifier generated by the Parichay application after successful user authentication which is used to identify user session per application.
Browser Id	Unique key to identify browser (set as a cookie in user's browser)
Encrypted String	The string contains user attributes (JSON format) encrypted with the service API key. Refer to Annexure C for User Attribute details.
Client Service Home URL	Home URL of the client service.
Client Signature	Hash generated using HMAC Algorithm
TTL	Time to live in milli seconds

BACKGROUND

Parichay is a single sign-on application designed to integrate NIC services under a single authentication domain. It is a centralized session and user authentication service in which one set of login credentials can be used to access multiple applications. The service authenticates user one on one designated platform, enabling the user to use a plethora of services without having to log in and logout each time. Once the user login to the Parichay, all the services that comply with Parichay's integration will be auto-logged in by sharing its session. It authenticates the user for all the services and does authorization based on the rights given to the user so it eliminates further authentication/authorization prompts when the user switches services during the same session.

INTRODUCTION

- Clients interested in availing the service of the e-Authentication framework should initially integrate their service with Parichay.
- Client Integration Framework enables Parichay to provide secure access to respective backend services through a well-defined gateway service that is consistent across all backend services, regardless of the service implementation.
- As a part of the framework, various services will be able to integrate with Parichay through offered Application Programming Interfaces (API) Web Service Interfaces smoothly and conveniently without affecting the existing architecture of the running services.

PRE-REQUISITE

- The client needs to submit the checklist mentioned in [Annexure A](#).
- Verify connectivity to Parichay application (Refer [Annexure B](#))
- The client service developers must have ample knowledge of their client service functionality (login/logout/timeout) and understanding of basic API functionality including knowledge of technology Jargons like Session/Cookies/Encryption-Decryption.

WHAT PARICHAY PROVIDES

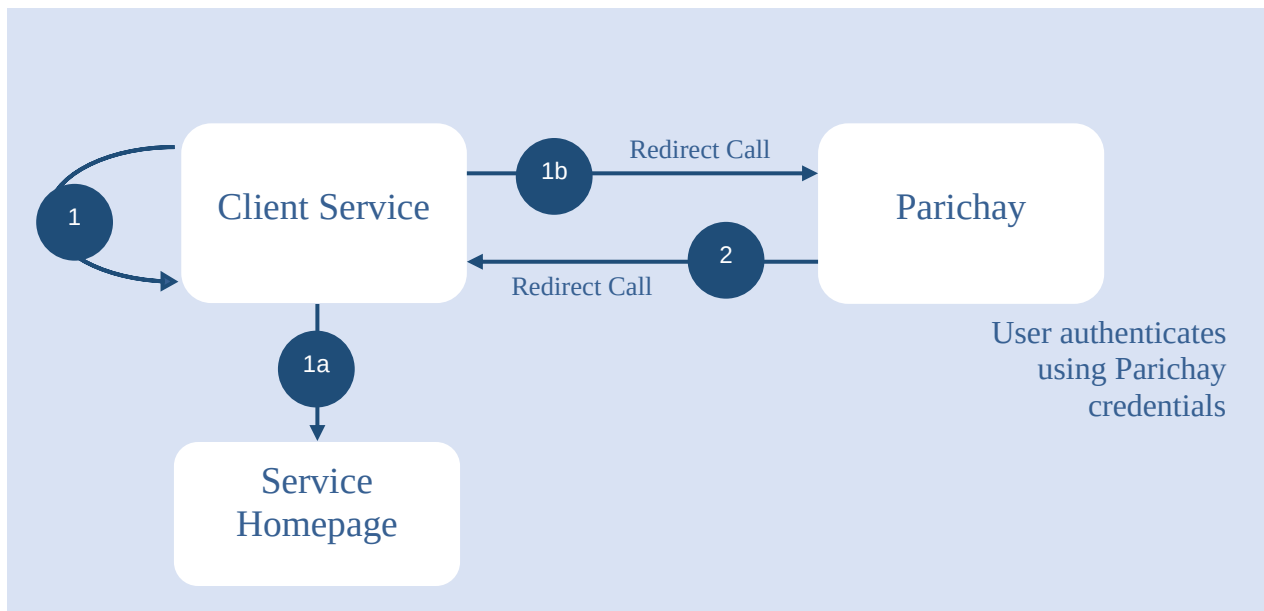
1. **Auth Key** - A 32 digit alphanumeric key generated per client application (client service) integrated with Parichay. The service API key is used to encrypt data shared between Parichay and the client application as well as used to generate Hash based message Authentication Code (HMAC).
2. **Rest Auth Id** - A 10 digit numeric key generated per client application (client service) integrated with Parichay. Rest Auth Id is used to map Service API key per client application.
3. **AES IV** - An initialization vector (IV) is an arbitrary number that can be used along with a secret key for data encryption.

PARICHAY INTEGRATION PROCESS

Client service needs to perform certain integration, which includes

- Login Process
- Logout Process
- Timeout Process
- Token Validation Process
- Handshaking Process

LOGIN PROCESS



1. When the user tries to access the client service (<https://vahan.nic.in/>), the service should check for **service session validation** by looking at the session cookie for the respective client service domain.
 - a) If the service **found** the required session cookies, the user should be redirected to the Service Home page.
 - b) If the service **does not found** the required session cookies or the session cookies found are already invalidated previously by the server, the client service web server should redirect the user to the Parichay URL (As mentioned below).

URL	https://PARICHAY_URL/pnv1/api/login?service=ServiceName&tid=TimeToLive&cs=ClientSignature&string=EncryptedClientId
HTTP Method	GET

Note

(A) *hmacString=“Parichay”+tid+https://PARICHAY_URL/pnv1/api/login+service*

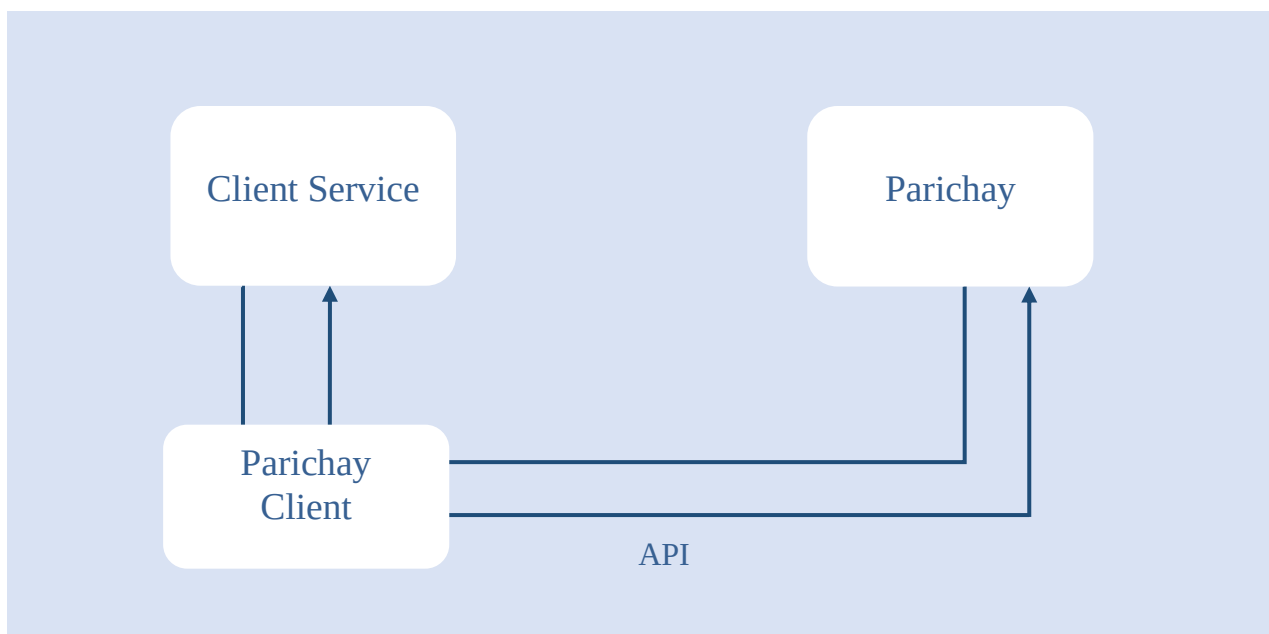
(B) *Client Session Id will be Encrypted (AES) using pre-shared Service API key*

(C) *Client must URL encode cs (Client Signature) and string (Encrypted Client Session Id)*

- If Parichay doesn't find any user session, the user will be asked for the credentials followed by the Two-step verification (if enabled) on the Parichay login page. After successful user authentication, Parichay calls HTTP redirect to the landing page of the registered client service (As mentioned below) with **Server Handshaking Id**.

URL	ClientServiceHomeURL?string=< Server Handshaking Id >
HTTP Method	GET

HANDSHAKING PROCESS



1. After receiving the Server Handshaking Id, the client service should take the string from the above URL and call the below-mentioned **Handshaking API**.

API	http://PARICHAY_CLIENT_URL/handshake OR http://PARICHAY_CLIENT_URL/Accounts/openam/login/validateClientToken
HTTP Method	GET/POST

Note: Port is configurable

Request Parameters

Attributes	Data Type	Required	Remarks
Server Handshaking Id	String	Yes	Path Parameter
Service Name	String	Yes	Path Parameter

Sample Request

http://PARICHAY_CLIENT_URL/handshake/ServerHandShakingId/ServiceName
OR
http://PARICHAY_CLIENT_URL/Accounts/openam/login/validateClientToken/
ServerHandShakingId/ServiceName

Note

(A) Success Response HTTP Status- 202 Accepted

(B) Server Handshaking Id used in the above URL is received from Parichay

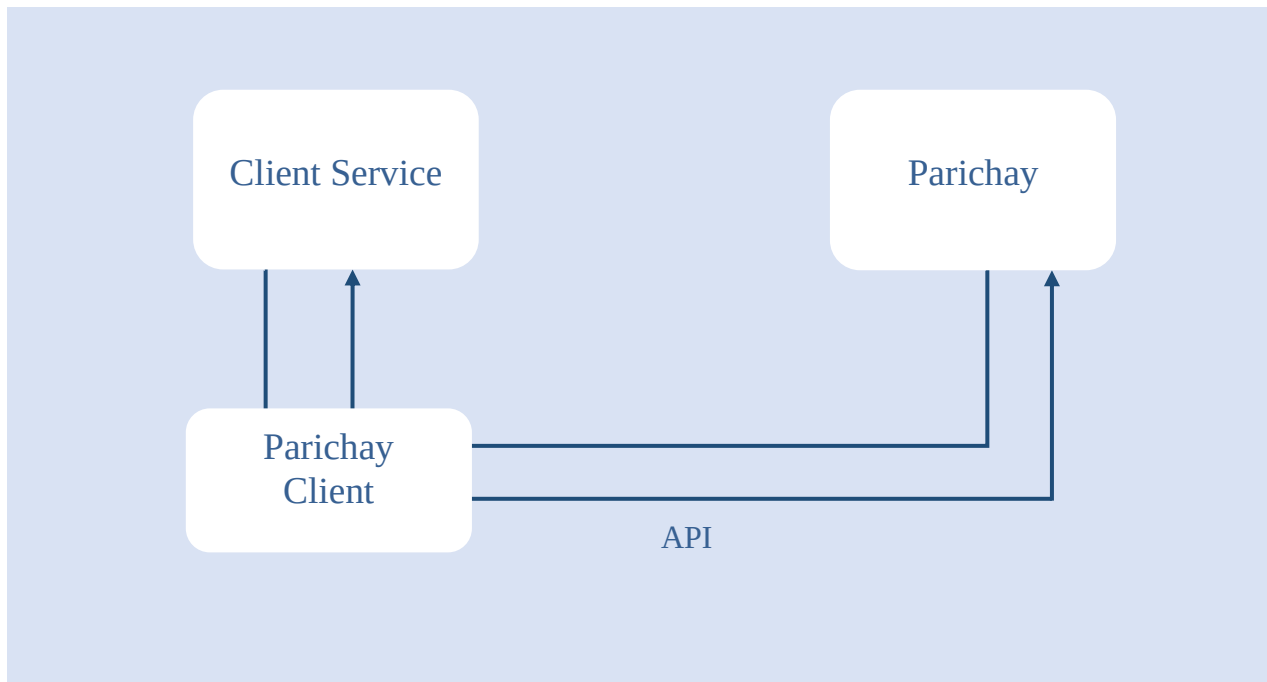
Response Payload Format

- a) Encrypted string, if string is found valid.
- b) False, if string is invalid.

2. The client service needs to decrypt the encrypted string received in response of Hand Shaking API with the Auth key provided to client service.
3. Now, Client service can start their session using the user parameters. (Refer to [Annexure C](#) for Parameters)

Note: String is decrypted with algorithm AESEncryption256 ECB

TOKEN VALIDATION PROCESS



1. Token validation is one of the mandatory API of the Parichay application as it **validates the user session** throughout the Parichay integrated services. If the user logout from any of the client service other client services of Parichay will not get to know that the client has been logged out until or unless the client service has performed the token validation.
2. Every time the user performs any functionality on client service, the client service must call the below mentioned Rest API to validate the user session.

API	http://PARICHAY_CLIENT_URL/isTokenValid OR http://PARICHAY_CLIENT_URL/Accounts/openam/login/isTokenValid
HTTP Method	GET/POST

Note: Port is configurable

Request Parameters

Attributes	Data Type	Required	Remarks
userName	String	Yes	Query Parameter
service	String	Yes	Query Parameter
sessionId	String	Yes	Query Parameter
browserId	String	Yes	Query Parameter
localTokenId	String	Yes	Query Parameter

Sample Request

```
http://PARICHAY_CLIENT_URL/isTokeValid?userName=test1@nic.in&service=test&sessionId=6B1F1ECD-624B-D701-2980-AF4DC93DEA60&browserId=125A72 DF-19FF-CED0-E084-25DB3A7B0655&localTokenId=AEJlu1844dhTyenIDMEKIjbEjebjJ
```

Note

(A) Success Response HTTP Status- 202 Accepted

Response Payload Format

Attributes	Remarks
status	API Response Status
tokenValid	Token Status

Sample Response Payload

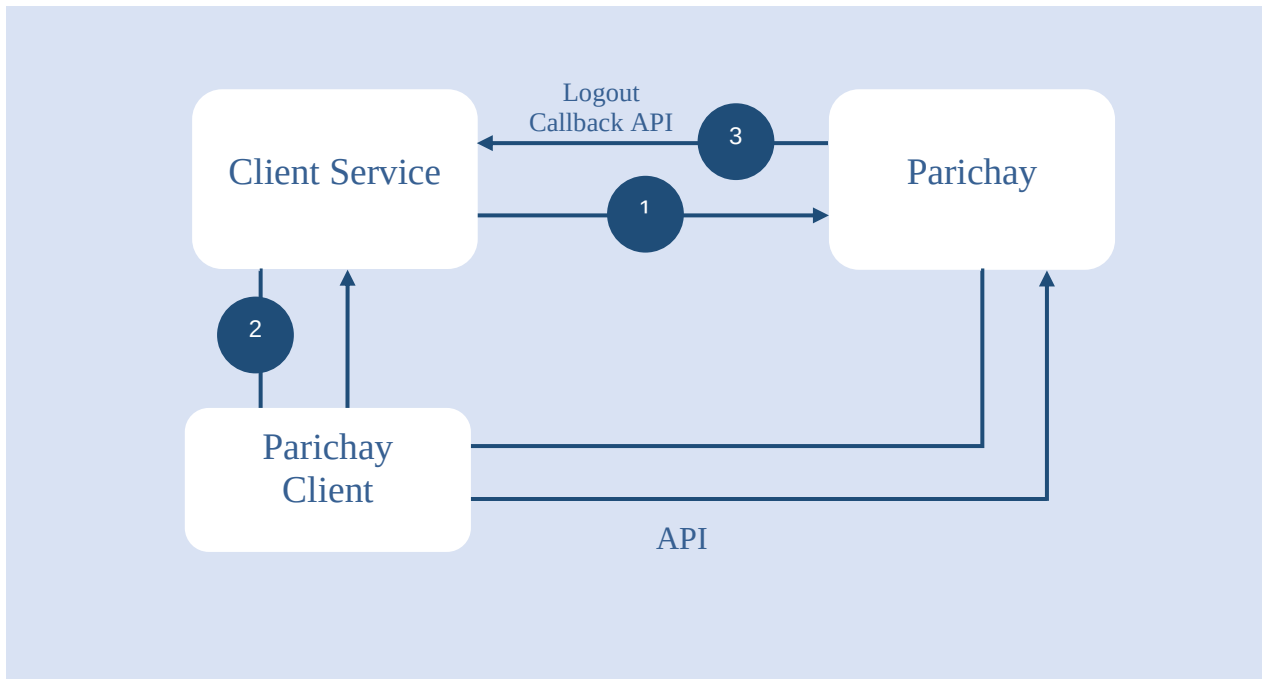
```
{  
  "status": "success", "tokenValid": "true"  
}
```

Note:

(A) Response status can be either "success" or "failure"

(B) "tokenValid" key value is either "true" or "false"

LOGOUT PROCESS



1. On clicking the Logout button on client service, the client service should first clear its session and call logout URL.

URL	<code>https://PARICHAY_URL/pnv1/salt/api/client/logout?userName=UserName&service=ServiceName&sessionId=SessionId&tid=TimeToLive&cs=ClientSignature</code>
HTTP Method	GET

Note

(A) $hmacString = "Parichay" + tid + "https://PARICHAY_URL/pnv1/salt/api/client/logout" + userName + service + sessionId$.

(B) If API is unreachable the client service should show the "Internal Server Error" page to the user instead of redirecting back to Parichay. In this case, refer to Parichay support for help.

(C) Logout from any service logs the user from all the services.

(D) Client must URL encode cs (Client Signature)

2. The client service may also choose to call Parichay logout API followed by re-direct URL.

API	http://PARICHAY_CLIENT_URL/logoutAll OR http://PARICHAY_CLIENT_URL/Accounts/openam/login/logoutAll
HTTP Method	POST

Note: Port is configurable

Request Parameters

Attributes	Data Type	Required	Remarks
Username	String	Yes	Message Body
Service	String	Yes	Message Body
sessionId	String	Yes	Message Body
browserId	String	Yes	Message Body
localTokenId	String	Yes	Message Body

Sample Request

```
http://PARICHAY_CLIENT_URL/logoutAll?userName=test1@nic.in&service=test&sessionId=6B1F1ECD-624B-D701-2980-AF4DC93DEA60&browserId=125A72 DF-19FF-CED0-E084-25DB3A7B0655&localTokenId=AEJlu1844dhTyenIDMEKljbEjebjJ
```

Note

(A) Success Response HTTP Status- 202 Accepted

Response Payload Format

Attributes	Remarks
Status	API response status
Message	Standard message

Sample Response Payload

```
{  
  "status": "success",  
  "message": "Respective  
message"  
}
```

Note:

(A) Response status can be either "success" or "failure"

3. To validate the user session throughout the integrated services, Parichay also uses **Logout Callback API** provided by the Client Service. It sets a callback function to inform all the integrated client services that the user has been logged out from one of the client services.
 - a) To use logout Callback API, client service needs to call the above Parichay logout API (provided in point 2).
 - b) Client should also create an API and expose it to Parichay. Through this API, Parichay will inform the client service to end their session as Parichay session has been already logged out.

Request Parameters

Attributes	Data Type	Required	Remarks
UserName	String	Yes	Message Body
ServiceName	String	Yes	Message Body
SessionId	String	Yes	Message Body
BrowserId	String	Yes	Message Body
ClientToken	String	Yes	Message Body

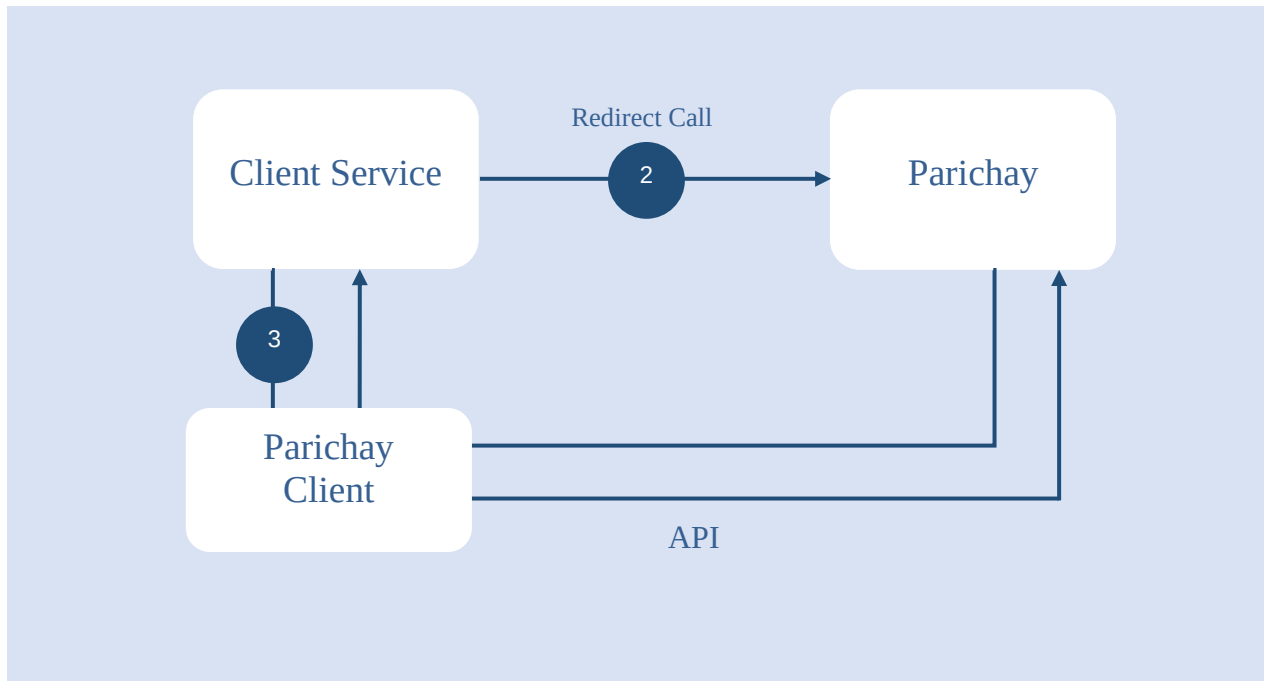
Sample Request Body

```
{"BrowserId":"","ClientToken":"","ServiceName":"","SessionId":"","UserName":""}
```

To know how to create service client API, kindly refer to [Annexure E](#).

Note: Clients may either use Logout Callback API or Token Validation API to validate the session.

TIMEOUT PROCESS



1. The session timeout for Parichay is 12 hours. However, each client service can also have its own session timeout.
2. If there is a session timeout of the client service, the client service should redirect the user to the Parichay password page via URL.

URL	<code>https://PARICHAY_URL/pnv1/salt/api/client/timeout?service=ServiceName&tid=TimeToLive&cs=ClientSignature</code>
HTTP Method	GET

Note

(A) $hmacString = "Parichay" + tid + "https://PARICHAY_URL/pnv1/salt/api/client/timeout" + service$.

(B) If API is unreachable the client service should show the "Internal Server Error" page to the user instead of redirecting back to Parichay. In this case, refer to Parichay support for help.

(C) Logout from any service logouts the user from all the services.

(D) Client must URL encode cs (Client Signature)

3. The client may also choose to call timeout API followed by re-direct URL.

API	http://PARICHAY_CLIENT_URL/timeout OR http://PARICHAY_CLIENT_URL/Accounts/openam/login/timeout
HTTP Method	POST

Request Parameters

Attributes	Data Type	Required	Remarks
userName	String	Yes	Message Body
service	String	Yes	Message Body
sessionId	String	Yes	Message Body
browserId	String	Yes	Message Body
localTokenId	String	Yes	Message Body

Sample Request

```
http://PARICHAY_CLIENT_URL/timeout?userName=test1@nic.in&service=test&sessionId=6B1F1ECD-624B-D701-2980-AF4DC93DEA60&browserId=125A72 DF-19FF-CED0-E084-25DB3A7B0655&localTokenId=AEJlu1844dhTyenIDMEKIjbEjebjJ
```

Note

(A) Success Response HTTP Status- 202 Accepted

Response Payload Format

Attributes	Remarks
status	API response status
message	Standard message

Sample Response Payload

```
{  
  "status": "success",  
  "message": "Respective  
message"
```

Note:

(A) Response status can be either "success" or "failure"

ADDITIONAL FRAMEWORK/API

HMAC GENERATOR

Hash-based Message Authentication Code (HMAC) generator uses Algorithms and secret key to generate the HMAC

API URL	http://PARICHAY_CLIENT_URL/hmac
HTTP Method	POST

Request Body

Attribute	Data Type	Required	Remarks
HmacString	String	Yes	Message Body

Note: Port is configurable

Request Body

```
{
  "HmacString":"Parichay18147832"
}
```

Note: Success Response HTTP Status- 202 Accepted

Response Payload Format

Attribute	Remarks
status	API response status
message	Standard message
Data	Response Data Object
data.signature	Generated HMAC Signature

Sample Response Payload

```
{
  "status": "success",
  "message": "Success message",
  "data": { "signature": "HMAC
            Sign" }
}
```

Note: Response status can be either "success" or "failure"

ENCRYPTION

Advanced Encryption Standard (AES) generator uses Algorithms with secret key and initialization vector to generate the encrypted string.

API URL	http://PARICHAY_CLIENT_URL/encryption
HTTP Method	POST

Note: Port is configurable

Response Payload Format

Attribute	Data Type	Required	Remarks
AESString	String	Yes	Message Body

Sample Request

```
{
  "AESString": "Parichay18147832"
}
```

Note: Success Response HTTP Status- 202 Accepted

Response Payload Format

Attribute	Remarks
status	API response status
message	Standard message
data	Response Data Object

data.signature	Generated Encrypted String
-----------------------	----------------------------

```
{  
  "status":"success",  
  "message":"Success message",  
  "data":{"signature":"Encrypted"  
}
```

Sample Response Payload

Note: Response status can be either "success" or "failure"

DECRYPTION

Advanced Encryption Standard (AES) generator uses Algorithms with secret key and initialization vector to decrypt the encrypted string.

API URL	http://PARICHAY_CLIENT_URL/decryption
HTTP Method	POST

Note: Port is configurable

Request Body

Attribute	Data Type	Required	Remarks
EncryptedString	String	Yes	Message Body

Sample Request

```
{
  "EncryptedString": "sahdbhbshsah"
}
```

Note: Success Response HTTP Status- 202 Accepted

Response Payload Format

Attribute	Remarks
Status	API response status
Message	Standard message
Data	Response Data Object
data.signature	Generated Decrypted String

Sample Response Payload

```
{
  "status": "success",
  "message": "Success message",
  "data": { "signature": "Decrypted" }
}
```

Note: Response status can be either "success" or "failure"

CLIENT FRAMEWORK IMPLEMENTATION

Client Framework is packaged for deployment to support both Linux and Windows environments. The following deployment packages are available,

1. Win64 Executable (ParichayClient.exe)
2. Linux Executable (ParichayClient)
3. Docker Image

The service is configured using the JSON configuration file (conf.json). The configuration parameters are given below:

Parameter	Description
ClientServicePort	The port on which the service is configured within the application/web server. This port must not be exposed outside the local server.
ClientServiceHost	The host of Client Integration Framework: localhost
ServiceName	Service Name assigned to client
Auth Key	Auth Key provided to client
AES IV	AES IV provided to client
Rest Auth Id	Rest Auth Id provided to client

Logging	Whether wants to enable logs or not
LogLevel	INFO/DEBUG

Note:

1. *Executable file and conf.json should be in the same directory*
2. *PARICHAY CLIENT URL formed using **ClientServiceHost:ClientServicePort** as mentioned in the above table*

ANNEXURE A

Requirements to Avail Parichay

Sr. No.	Requirements
1.	Name of the Service *Name of the client application
2.	Single Login/Multiple Login *Single login services can be accessed only in Single browser at a time while multiple login services can be accessed in multiple browsers (e.g. Chrome and Mozilla) at a time
3.	Enforce Multi-Factor Authentication Yes/No *Should all users enforced for Multi-factor authentication by client service or not.
4.	Force Multi-Factor Authentication Yes/No *Should all users forced for Multi-factor authentication by client service or not
5.	Service Description *A small description of the service which will be shown on the hover of the application logo
6.	Department Name *Name of the Department the application belongs
7.	Client Service Home URL *Landing URL of the client application. User will be redirected back to this URL after successful authentication from Parichay
8.	Client Service Login URL *Login page URL of the client service. User will be redirected to this URL in case of any error while accessing the service
9.	Client Service Logout URL *A re-direct Logout URL of the client service
10.	Enforce Multi-Factor Authentication per user (If yes, mention user details as per below format) Primary mail Id, Role (Admin/HR/Department Head/Manager/Other), Multi-Factor Authentication (Yes/No) * Should a user enforced for Multi-factor Authentication by client service or not
11.	Select Environment <u>Staging/ Pre-Production/ Production</u>
12.	Is the application enabled over https Yes/No

13.	Is the application security audit cleared (Only for Production Environment) Yes/No If yes, share the Security Audit Certificate
14.	Enforce Geofencing Geofencing per user: Yes/No Geofencing per service: Yes/No * Should a service restricts access to a particular state or a country
15.	Service owner details • Name • Email • Mobile
16.	Service State

Note: In case of adding any new parameter, kindly contact Parichay support (Refer [Annexure D](#)).

ANNEXURE B

Verify Connectivity with Parichay

Parichay Environment	Parichay Domain (PARICHAY URL)	Ports	Connectivity
Staging	parichay.staging.nic.in	443	telnet parichay.staging.nic.in 443
Pre-Production	parichay.pp.nic.in	443	telnet parichay.pp.nic.in 443
Production	parichay.nic.in	443	telnet parichay.nic.in 443

Note: In case, IP or Port doesn't respond, ask Network or Firewall team to check respectively.

ANNEXURE C

User Attribute details in JSON

Key	Value
firstName	First Name
lastName	Last Name
email	Email
mobileNo	Mobile No
designation	Designation
address	Address
status	User Account Status
userId	User Id
parichayId	Parichay Id
ip	IP
browserId	Browser Id
ua	User Agent
parentToken	Post Login Session Id
clientToken	Client Token
loginId	Login Id
verificationParameters	User Role (If service has required)

Note: In case of adding any new parameter, kindly contact Parichay support (Refer Annexure D).

ANNEXURE D

Parichay Support

Contact Number	011-22902494
VoIP	7494
Email Id	support-Parichay@nic.in , support.perf@nkn.in

ANNEXURE E

How to create Service Client API

Step 1: Fetch following keys from POST parameters:

- a) UserName
- b) ServiceName
- c) SessionId
- d) ClientToken
- e) BrowserId

Example: {"BrowserId":"","ClientToken":"","ServiceName":"","SessionId":"","UserName":""}

Step 2: Fetch following parameters from request headers:

- a) Signature
- b) TTL

Step 3: Calculate HMAC from POST Data and Header Data, refer Annexure A.

Step 4: Match the Client Hash with Server Signature:

- a) If not matched, return "FAILURE" as API response.
- b) If matched, return "SUCCESS" as API response.

HMAC Calculation (Sample Data)

- <ttl> is current timestamp in milliseconds.
- <Signature> is base64 encoded form of Timestamp, Logout URL, and Body Param Values (all values concatenated)

For example, following is concatenated string to create signature:

160553336176<http://parichaytesting1.dev.nic.in/logouttest.php>**8076123F37A50E**
5D7F2447BDA702BA02B00C843B698844CCC747C51C08767E41E46C3C9D7E3A1892

6FF7CD1EBD89DEE28B141627DC39BB3E8BCAF9D09C5461CAservice1user.nknsp@nic.in7B4D5923-7033-4439-C197-5FEB341224F48760AC52-AB5D-B11B-6F02-2829260D445D

Here,

- 1605533336176 is **Current Timestamp in microseconds (ttl)**.
- http://parichaytesting1.dev.nic.in/logouttest.php is **Logout API URL**.
- 8076123F37A50E5D7F2447BDA702BA02B00C843B698844CCC747C51C08767E41E463C9D7E3A18926FF7CD1EBD89DEE28B141627DC39BB3E8BCAF9D09C5461CA is **Client Token**
- service1 is **Servicename**
- user.nknsp@nic.in is **Username**
- 7B4D5923-7033-4439-C197-5FEB341224F4 is **SessionID**
- 8760AC52-AB5D-B11B-6F02-2829260D445D is **BrowserID**

8076123F37A50E5D7F2447BDA702BA02B00C843B698844CCC747C51C08767E41E463C9D7E3A18926FF7CD1EBD89DEE28B141627DC39BB3E8BCAF9D09C5461CAservice1user.nknsp@nic.in7B4D5923-7033-4439-C197-5FEB341224F48760AC52-AB5D-B11B-6F02-2829260D445D is Final Post Body for HMAC. (Maintain values sequence for proper HMAC Calculation)

Prepare HMAC of above data (i.e. of sample signature) using pre shared Secret Key.
HMAC Algorithm used: sha512